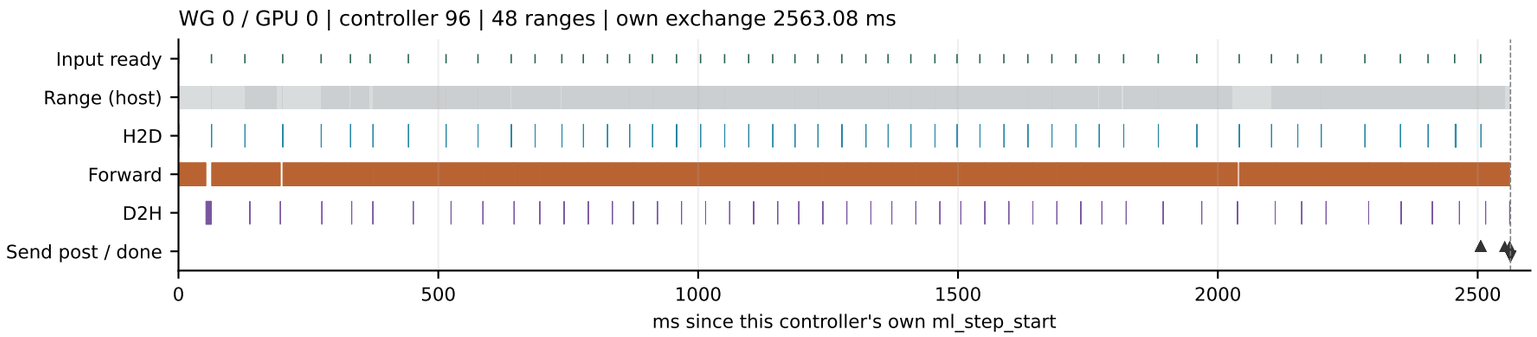
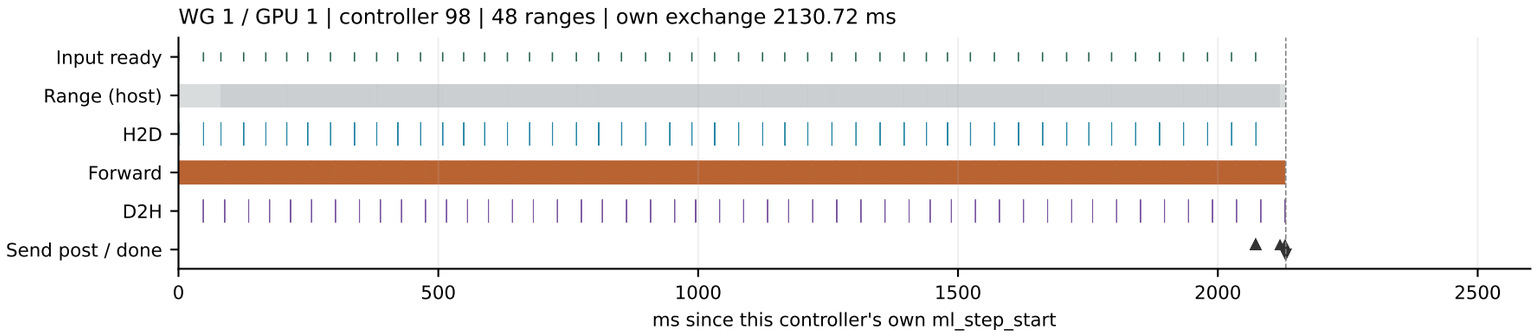
transformer | 96p96c4g | numa | collective vs p2pfull | rep 1 | call 6 / MAIA step 189

Overall MAIA ML-step makespan: collective 1748.16 ms | p2pfull 2589.61 ms



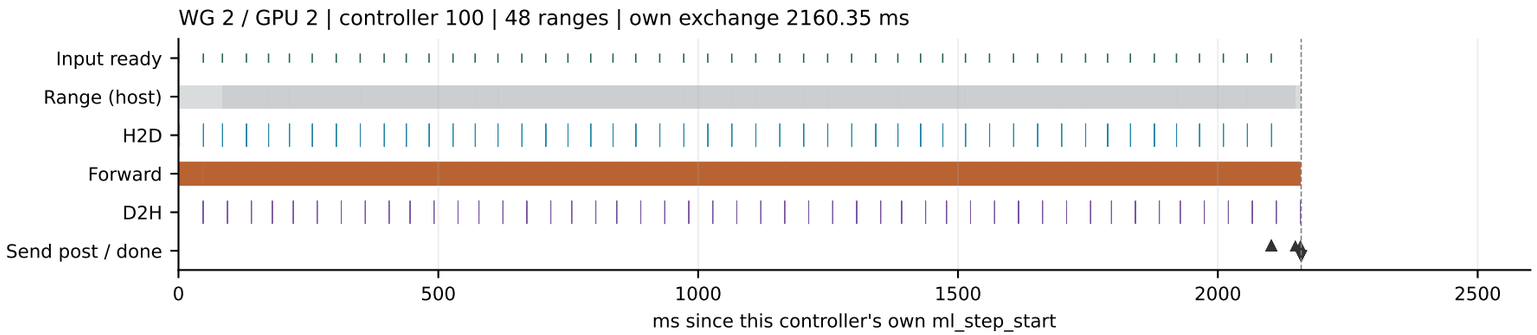
GPU work (ms)	Collective	p2pfull
H2D	5.401	17.618
Forward	1665.057	2540.150
D2H	7.200	18.058
GPU SUM	1677.658	2575.826

Controllers: coll 96 / P2P 96
Coll host device elapsed: 1678.708 ms
P2P GPU envelope: 2561.368 ms
Different scopes; not an elapsed-time comparison.



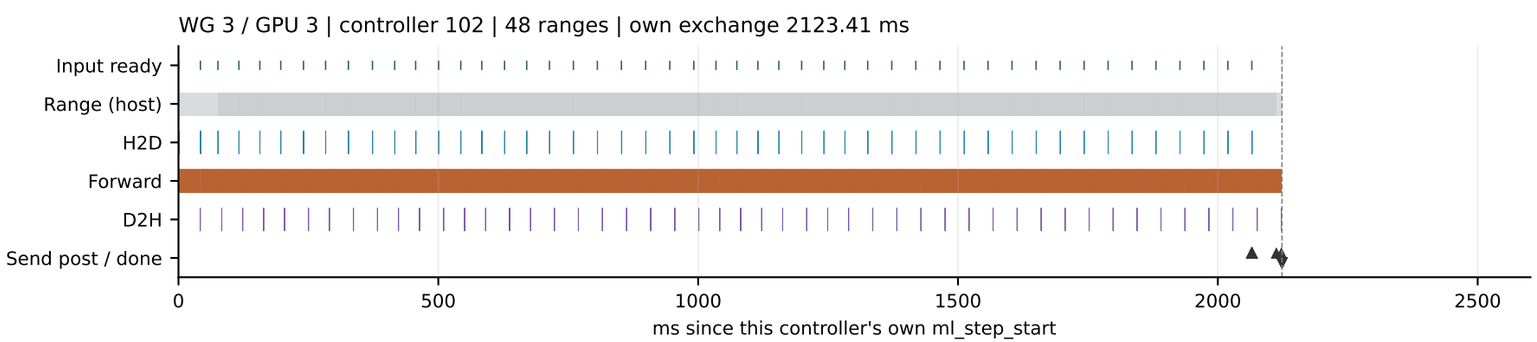
GPU work (ms)	Collective	p2pfull
H2D	5.779	17.956
Forward	1617.951	2127.432
D2H	6.283	7.447
GPU SUM	1630.013	2152.834

Controllers: coll 98 / P2P 98
Coll host device elapsed: 1630.566 ms
P2P GPU envelope: 2129.122 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2pfull
H2D	5.097	15.327
Forward	1657.789	2156.847
D2H	5.643	7.739
GPU SUM	1668.528	2179.913

Controllers: coll 100 / P2P 100
Coll host device elapsed: 1669.044 ms
P2P GPU envelope: 2158.637 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2pfull
H2D	5.159	18.371
Forward	1607.614	2119.653
D2H	4.607	7.673
GPU SUM	1617.380	2145.697

Controllers: coll 102 / P2P 102
Coll host device elapsed: 1618.148 ms
P2P GPU envelope: 2121.670 ms
Different scopes; not an elapsed-time comparison.

Each controller has its own zero; rows are NOT globally synchronized. CUDA-event durations are reliable; host-anchored inter-request overlap is approximate. GPU work SUM is not elapsed time or a critical path.